



Transferable Graph Learning for Transmission Congestion Management via Busbar Splitting

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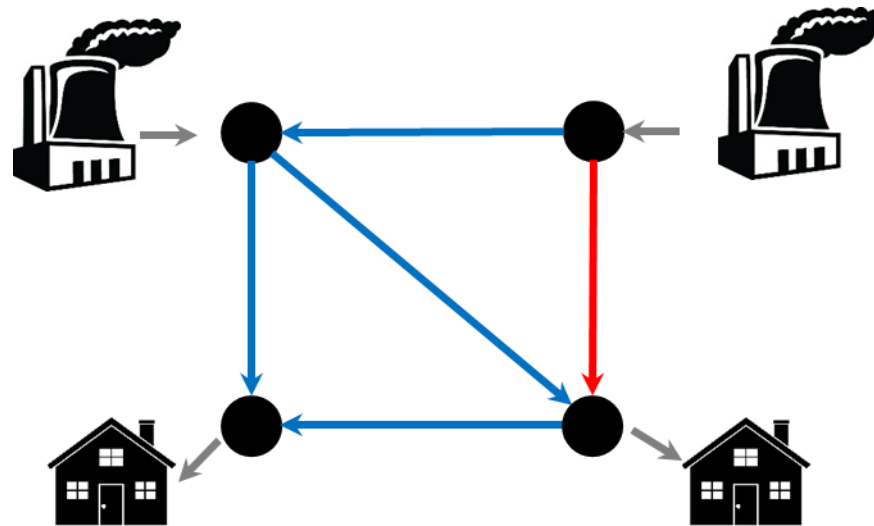
Motivation

- Congestion



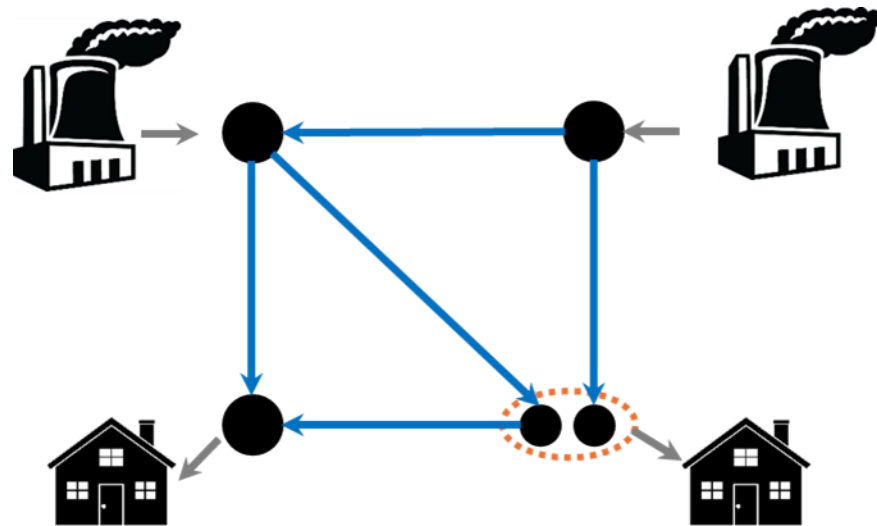
Motivation

- Congestion
- Topology optimization



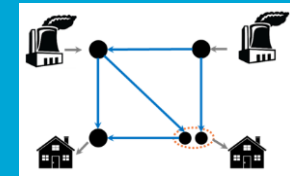
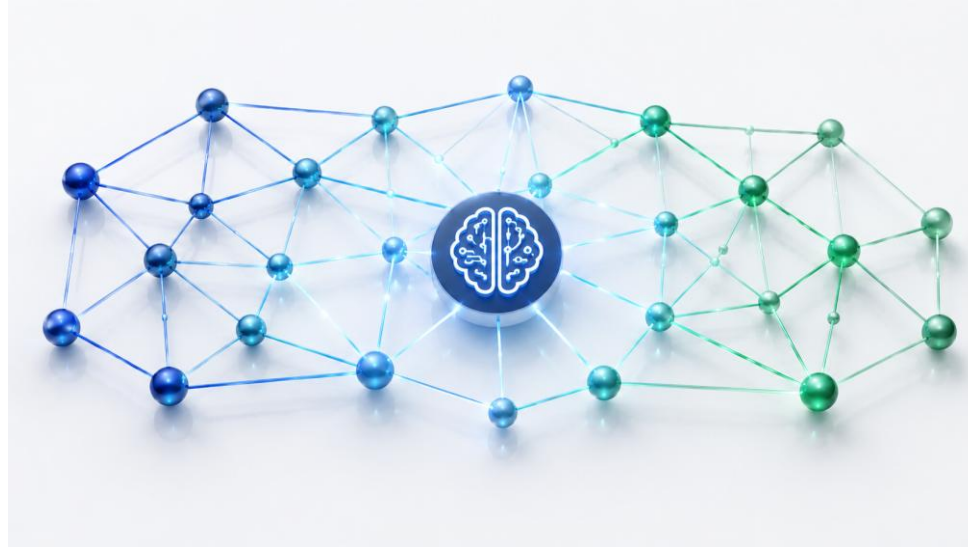
Motivation

- Congestion
- Topology optimization



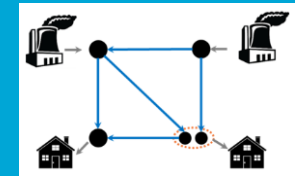
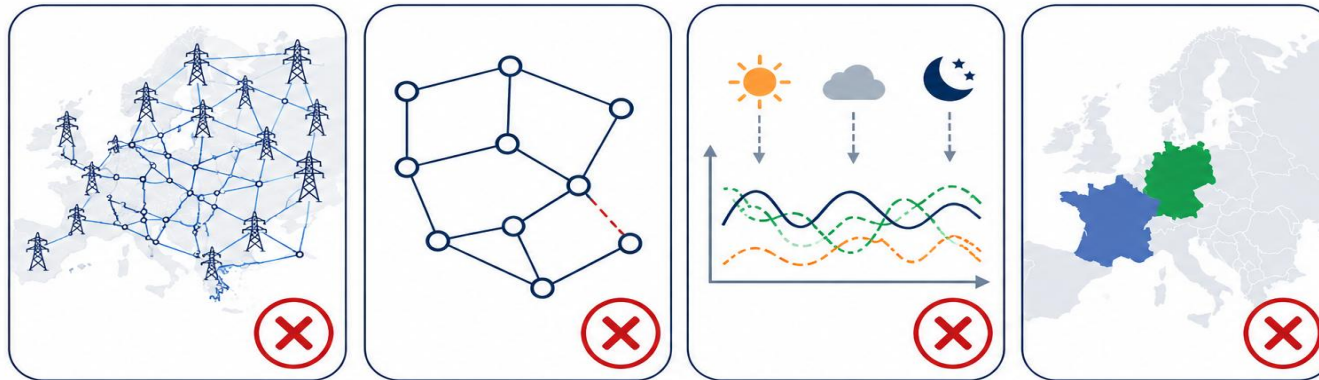
Motivation

- Congestion
- Topology optimization
- Machine Learning



Motivation

- Congestion
- Topology optimization
- Machine Learning



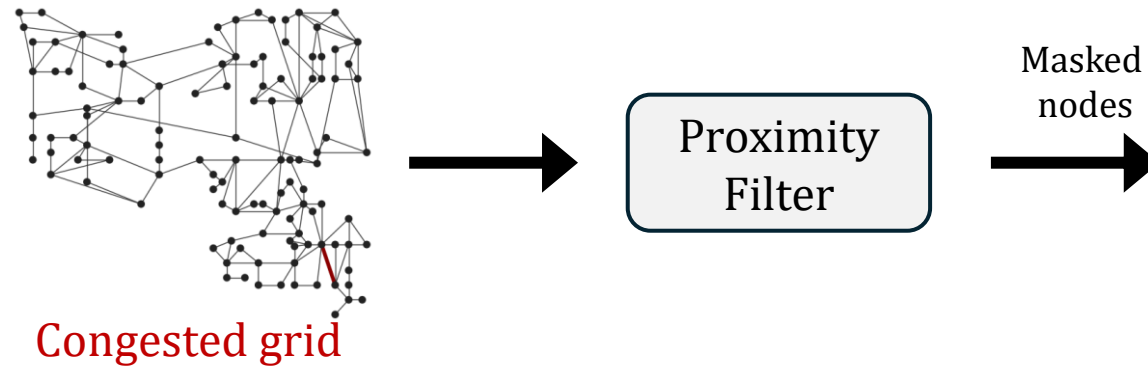
Transferable graph learning
to find fast busbar splitting actions
that reduce congestion.

Workflow

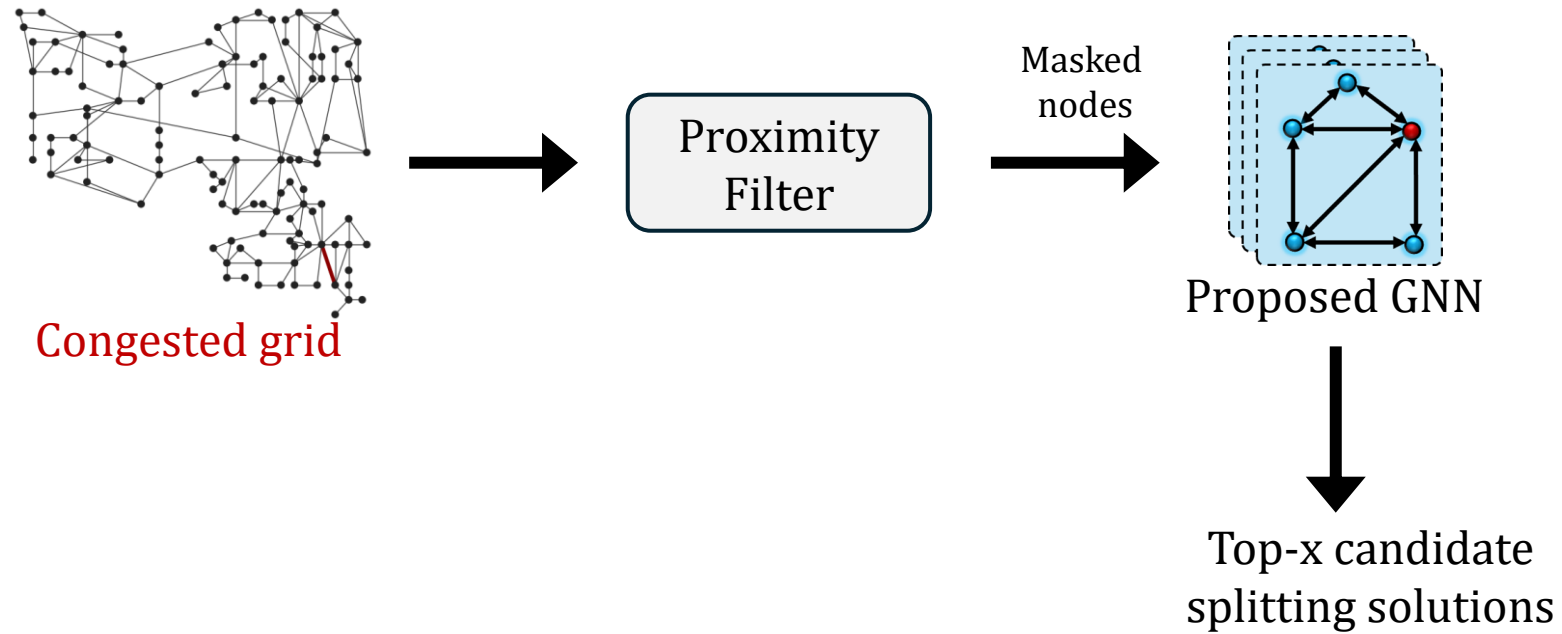


Congested grid

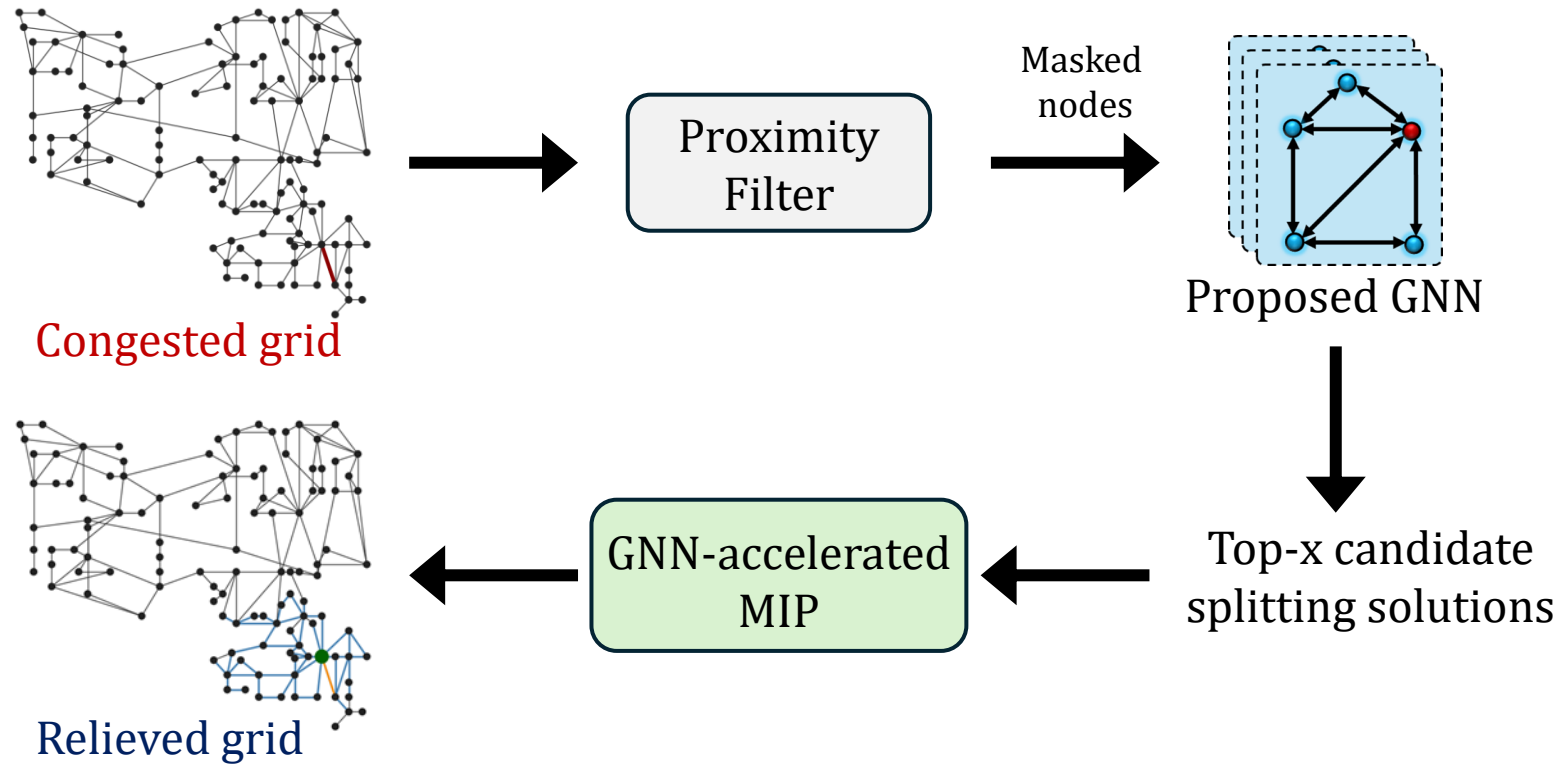
Workflow



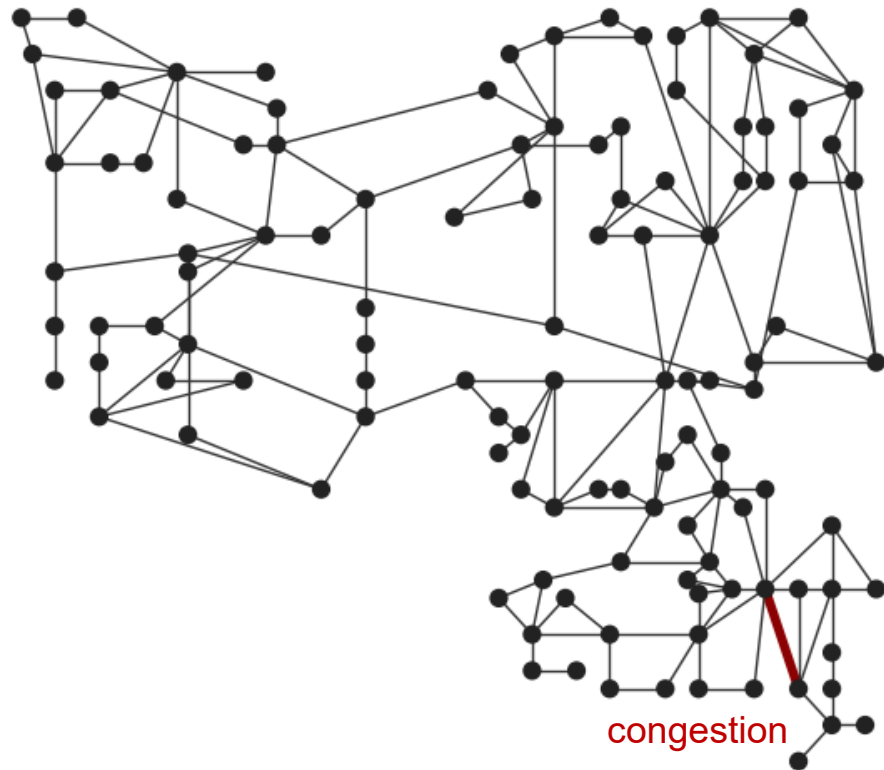
Workflow



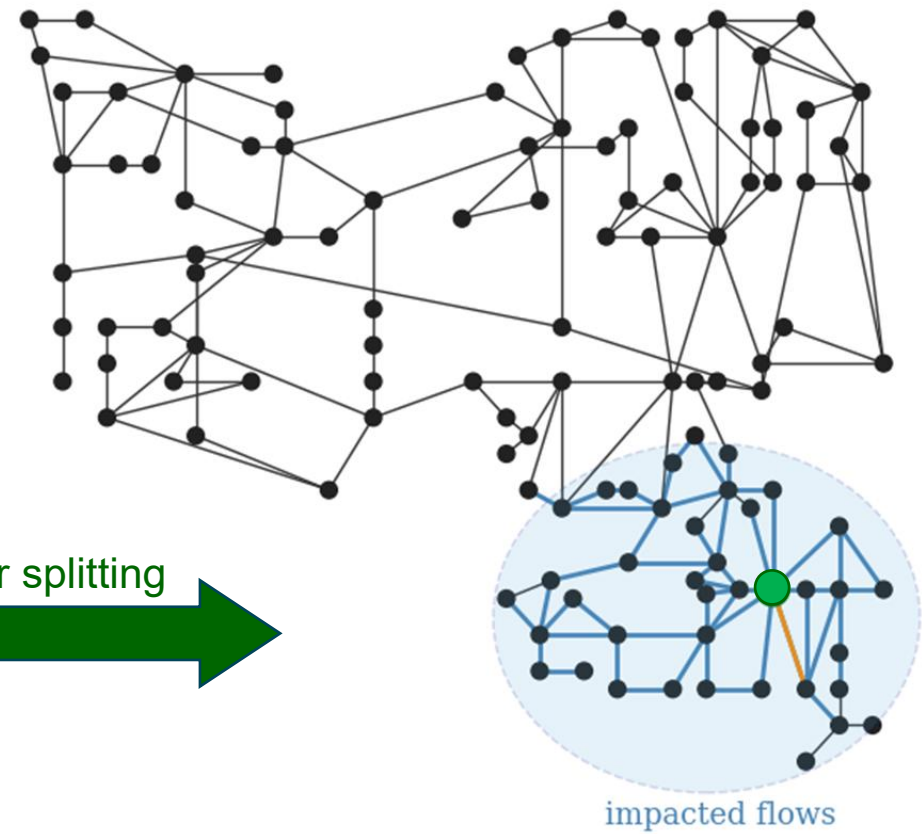
Workflow



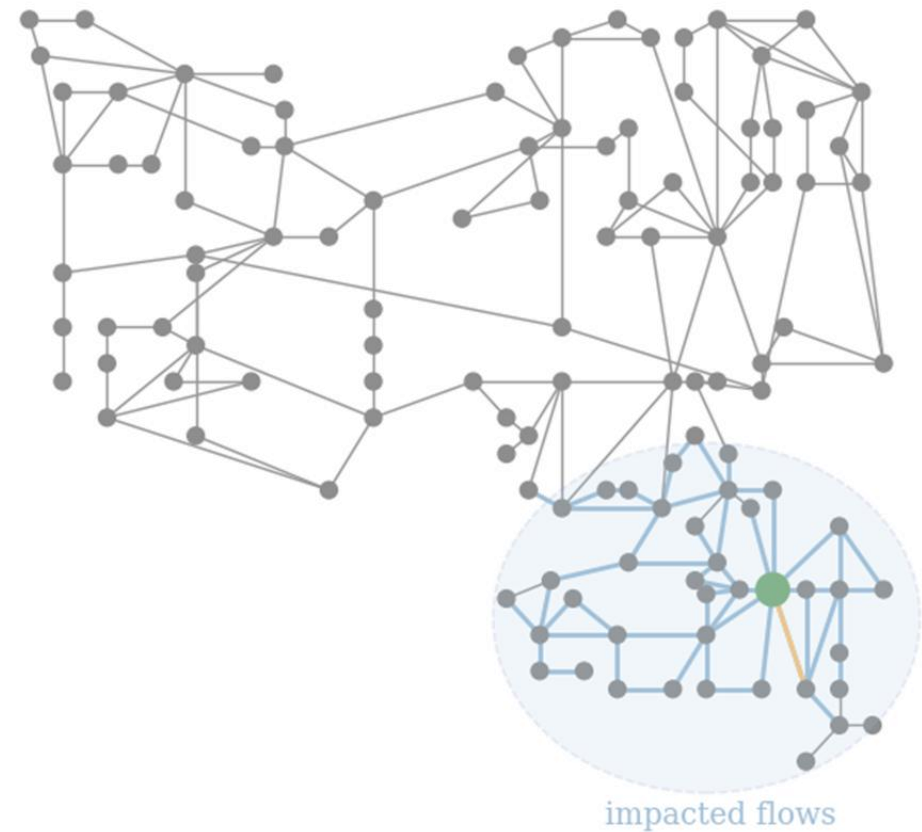
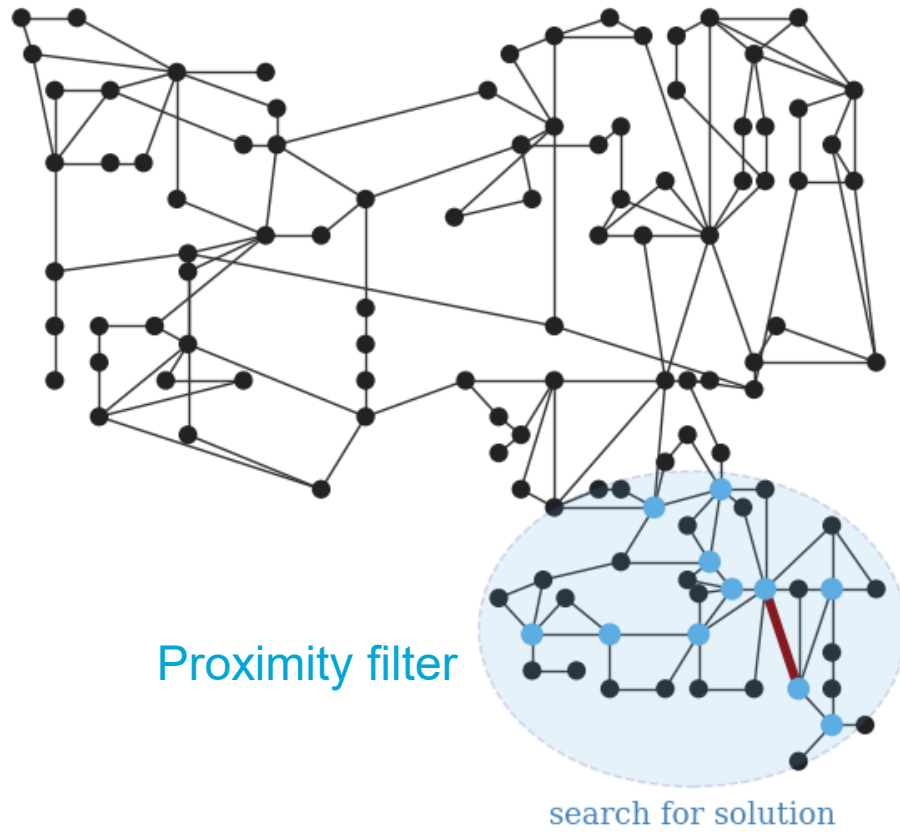
Proximity



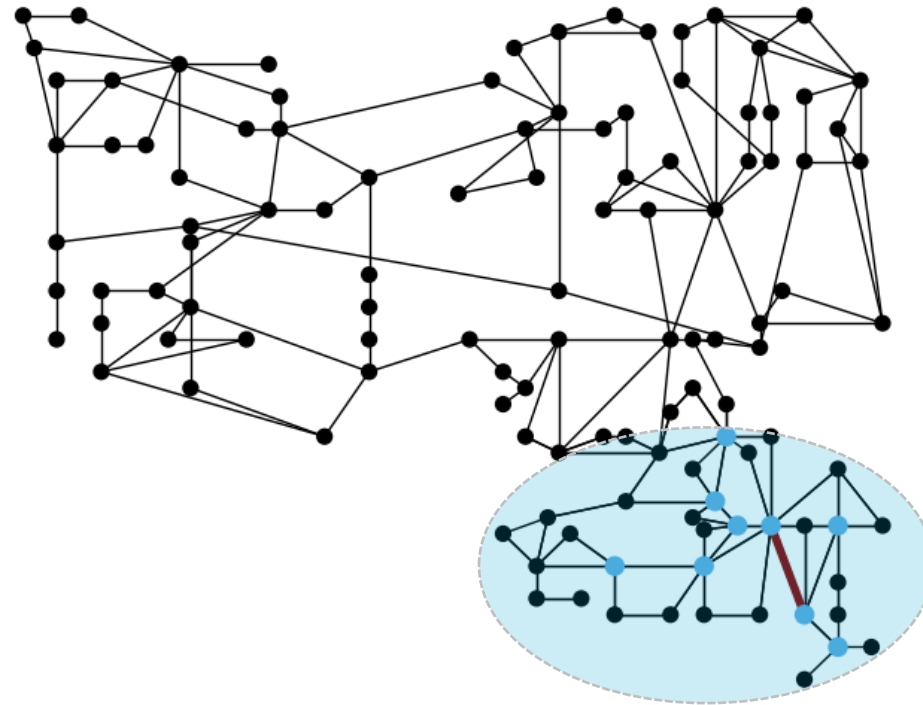
busbar splitting



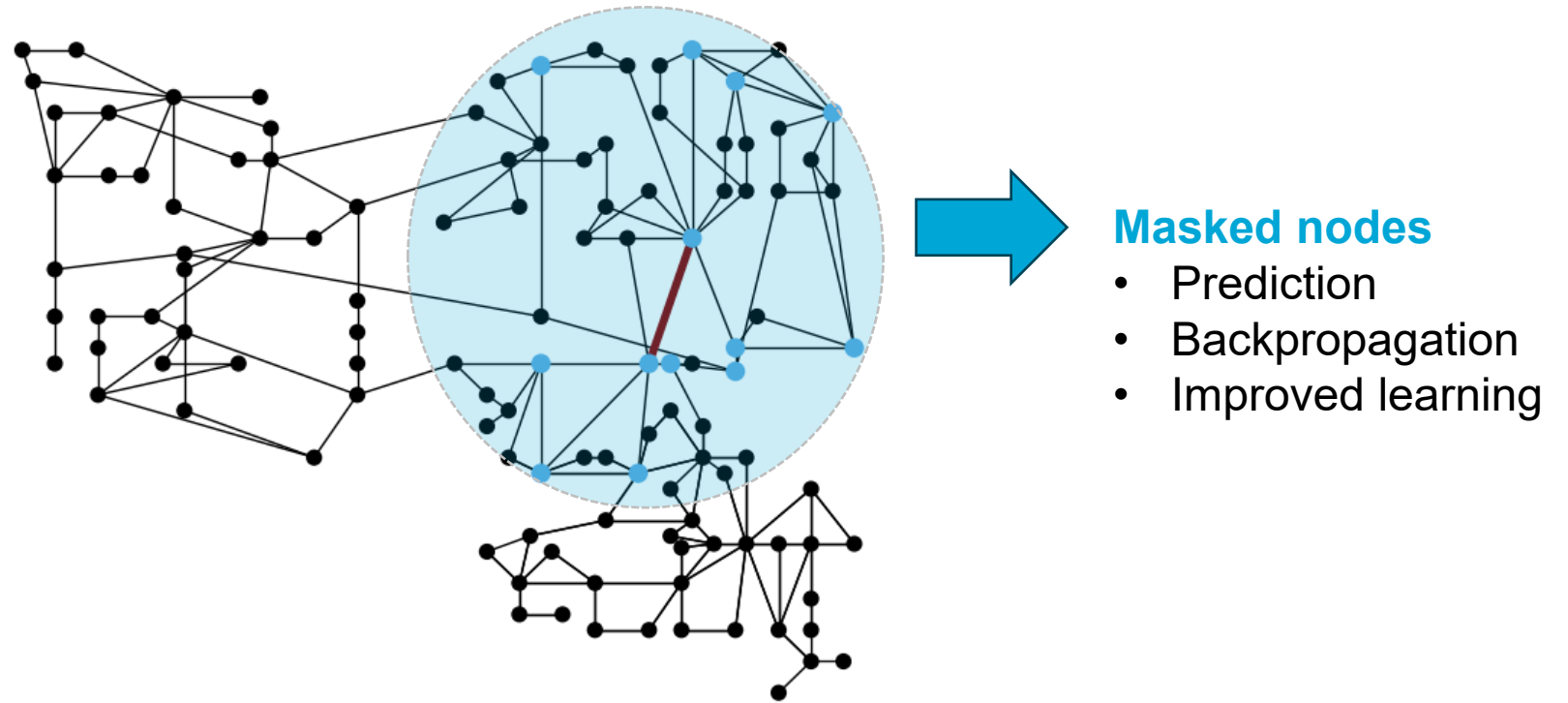
Proximity



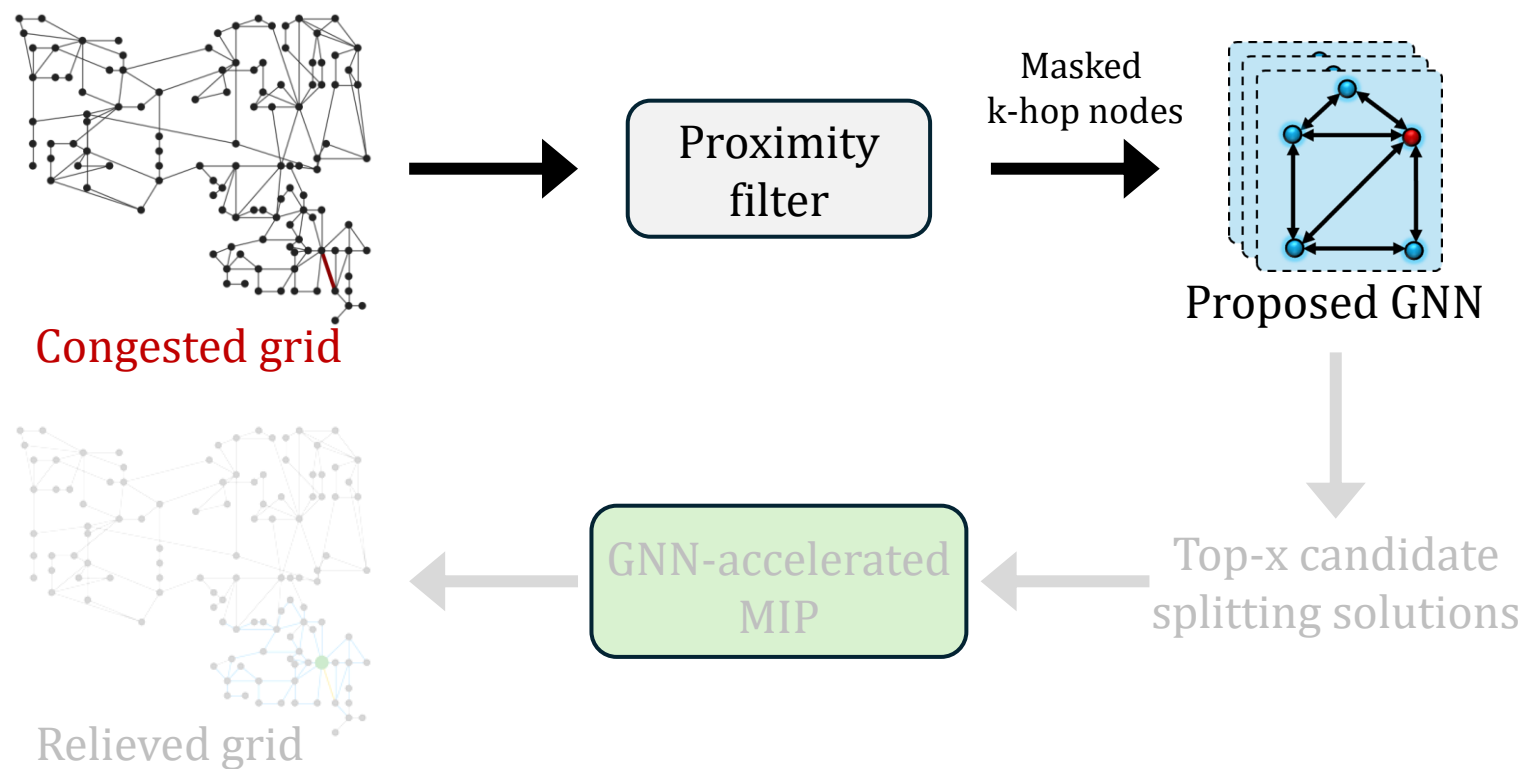
Proximity Filter



Proximity Filter



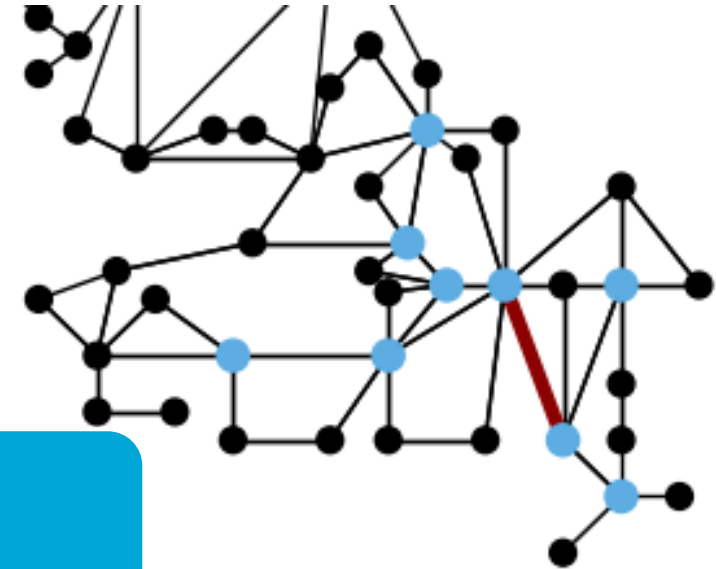
Workflow



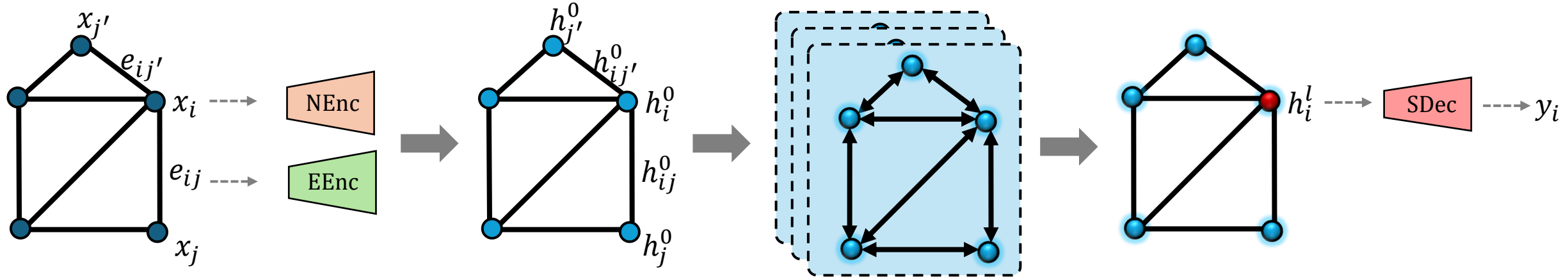
Proposed GNN: Learning Task

- Node classification:
 - $y_i \in \{0,1\}$: *whether to split node i*
 - *Note: not the full substation configuration*
- Node regression:
 - $y_i \in \mathbb{R}$: *how much congestion*

What is the GNN output?



Proposed GNN: Architecture



- $x_i = [p_i, q_i, V_i, \text{Deg}_i, \text{Dis}_i]$
- $e_{ij} = [P_{l,i}, Q_{l,i}, S_{l,i}, \bar{S}_l, r_l, x_l]$

Message passing neural network (MPNN)

$$h_i^0 = \phi^{NEnc}(x_i) \quad i \in \mathcal{V} \quad (7a)$$

$$h_{i,j}^0 = \phi^{EEnc}(e_{i,j}) \quad ij \in \mathcal{L} \quad (7b)$$

$$\tilde{h}_{i,j}^l = \phi_{\text{msg}}^l([h_i^{l-1} || h_j^{l-1} || h_{i,j}^{l-1}]) \quad ij \in \mathcal{L} \quad (7c)$$

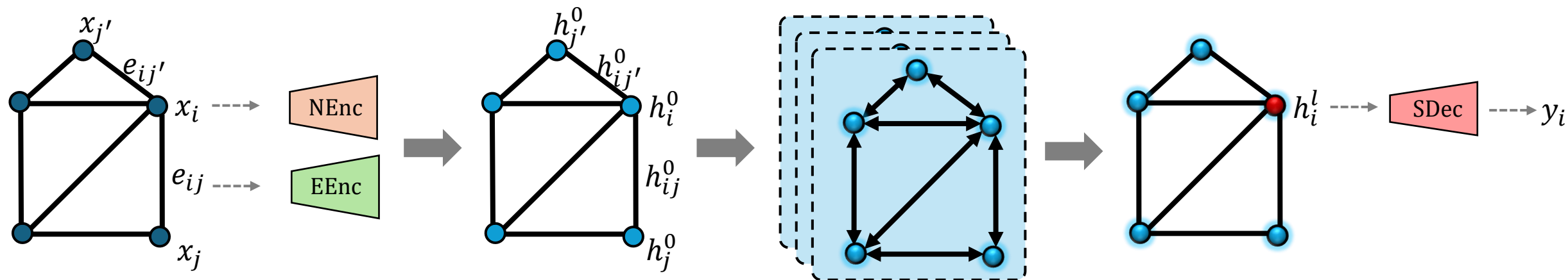
$$h_{i,j}^l = h_{i,j}^{l-1} + \tilde{h}_{i,j}^l \quad ij \in \mathcal{L} \quad (7d)$$

$$m_i^l = \sum_{j \in N_i} (h_{i,j}^l) \quad i \in \mathcal{V} \quad (7e)$$

$$h_i^l = h_i^{l-1} + \phi_{\text{upd}}^l([h_i^{l-1} || m_i^l]) \quad i \in \mathcal{V} \quad (7f)$$

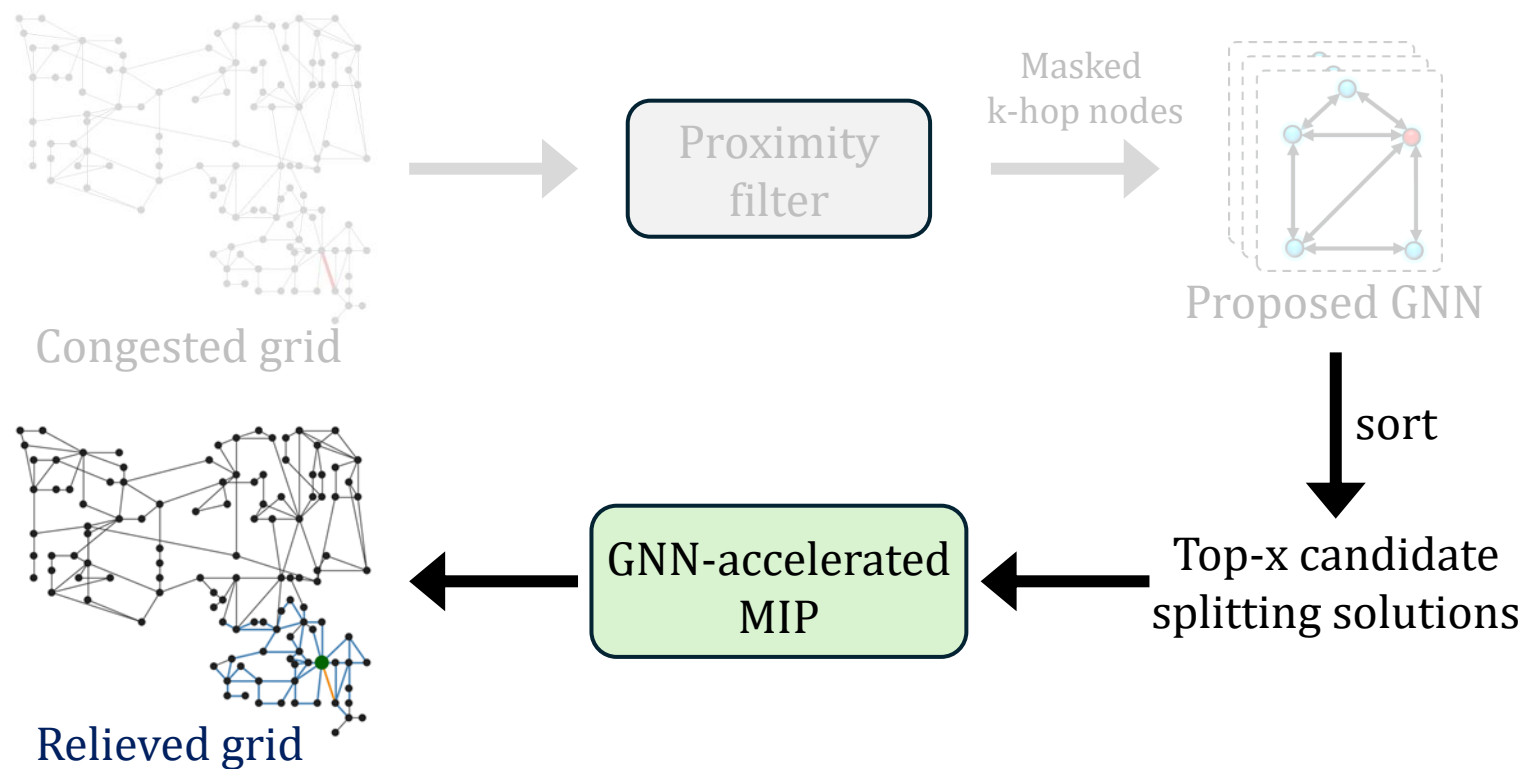
$$\hat{y}_i = \phi^{SDec}(h_i^L) \quad i \in \mathcal{V}^p \quad (7g)$$

Proposed GNN: Architecture

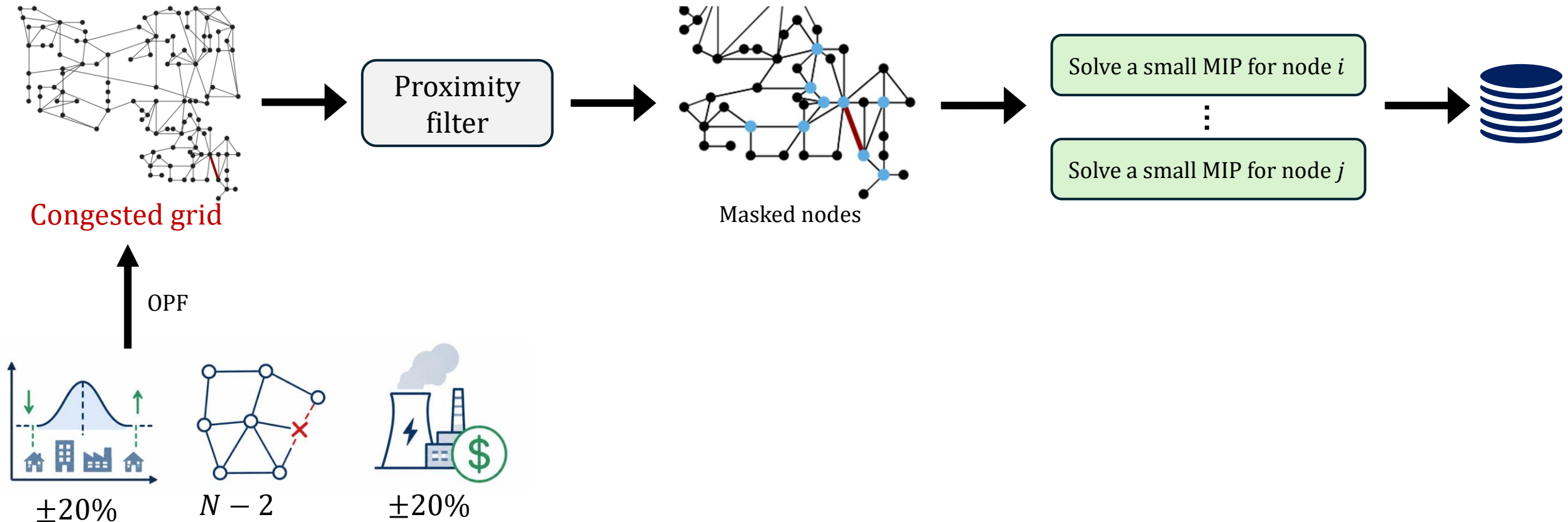


- Homogeneous
- Heterogeneous:
 - splitting-nodes (masked nodes)
 - non-splitting-nodes (not masked nodes)

Workflow

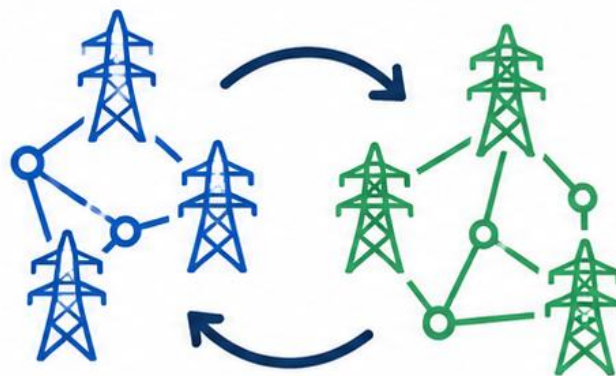
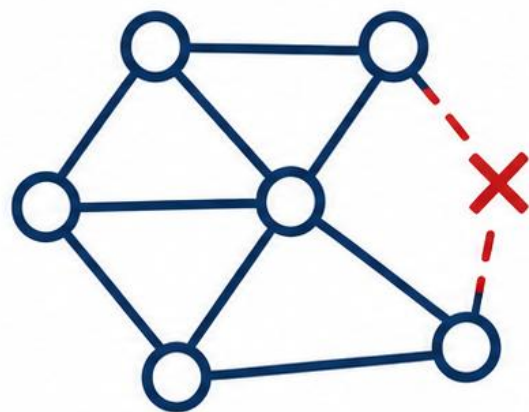


Training Data Generation

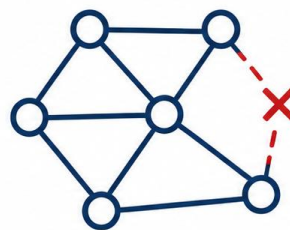


Case studies

- IEEE-118, IEEE-300, GOC-500, GOC-2000
- Models:
 - Multi-layer perceptron (MLP)
 - Graph convolution network (GCN)
 - Homo- & Het-MPNN



Generalization to Topology Changes

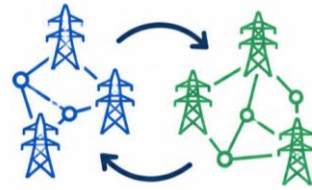


- Performance drop from $N-1 \rightarrow N-2$ is lower than $N-0 \rightarrow N-1$
- Combined dataset ✓
- Het-GNN ✗

TABLE II: Generalization to N-k topology changes.

	Train set	N-0	N-0	N-1	N-1	N-0, N-1, N-2
	Test set	N-0	N-1	N-1	N-2	N-0, N-1, N-2
ieee-14	MLP	0.97	0.79	0.93	0.79	0.92
	Hom-GCN	0.97	0.70	0.92	0.77	0.93
	Hom-MPNN	0.98	0.80	0.95	0.85	0.96
	Het-MPNN	0.98	0.81	0.96	0.86	0.96
ieee-118	MLP	0.87	0.79	0.82	0.79	0.83
	Hom-GCN	0.84	0.80	0.80	0.78	0.82
	Hom-MPNN	0.90	0.86	0.87	0.86	0.89
	Het-MPNN	0.91	0.88	0.88	0.87	0.90
ieee-300	MLP	0.87	0.85	0.85	0.84	0.87
	Hom-GCN	0.78	0.78	0.78	0.77	0.80
	Hom-MPNN	0.91	0.90	0.89	0.88	0.90
	Het-MPNN	0.91	0.90	0.89	0.88	0.91

Transferability across Systems



- Zero-shot ✗
- Transfer learning ✓

TABLE III: Transferability across systems

Source System	Target System		Zero-Shot	Transfer Learning	Combined Dataset
ieee-14	ieee-118	MLP	-	-	-
		Hom-GCN	0.39	0.78	0.85
		Hom-MPNN	0.28	0.86	0.90
		Het-MPNN	0.30	0.86	0.90
ieee-118	ieee-300	MLP			
		Hom-GCN	0.28	0.77	0.80
		Hom-MPNN	0.39	0.87	0.90
		Het-MPNN	0.31	0.87	0.91
goc-500	goc-2000	MLP			
		Hom-GCN	0.32	0.83	0.85
		Hom-MPNN	0.30	0.91	0.93
		Het-MPNN	0.28	0.92	0.95

Computational Efficiency



- Speed-up $\times 10^2 - 10^3$
- Optimality gap:
 - 1-split ✓
 - 3-split ✗
- Reg vs Clf

TABLE IV: Computational efficiency of different NTO-CM approaches.

		DC 1-split			DC 3-split			AC 1-split		
		Time	Speed up (x)	Gap (%)	Time	Speed up (x)	Gap (%)	Time	speed up (x)	gap (%)
ieee-118	Org-NTD	18.3 s	-	-	2.2 min	1x	0.0	1.5 min	-	-
	5-hop-NTD	18.3 s	1x	0.4	45.7 s	3x	0.8	9.5 s	10x	0.5
	No-SW	0.4 s	46x	8.9	0.5 s	283x	23.1	0.6 s	149x	6.7
	Clf-GNN-NTD	0.9 s	21x	2.2	5.2 s	26x	7.3	2.7 s	34x	3.3
	Reg-GNN-NTD	1.3 s	14x	1.7	10.0 s	13x	2.2	3.2 s	29x	1.2
ieee-300	Org-NTD	49.8 s	-	-	12.1 min	1x	0.0	3.4 min	-	-
	5-hop-NTD	24.0 s	2x	0.4	8.1 min	1x	0.7	2.0 min	2x	0.2
	No-SW	0.6 s	84x	10.2	0.8 s	949x	41.1	1.5 s	140x	1.4
	Clf-GNN-NTD	1.5 s	33x	5.5	6.6 s	110x	12.1	4.0 s	51x	0.4
	Reg-GNN-NTD	2.1 s	24x	1.1	22.9 s	32x	6.0	9.3 s	22x	0.2
goc-500	Org-NTD	4.1 min	-	-	35.6 min	1x	0.0	46.7 min	-	-
	5-hop-NTD	2.1 min	2x	0.8	5.9 min	6x	3.2	17.2 min	3x	0.0
	No-SW	1.1 s	230x	2.5	1.2 s	1838x	10.6	3.9 s	712x	1.7
	Clf-GNN-NTD	2.3 s	108x	1.9	11.0 s	194x	5.1	8.9 s	314x	2.6
	Reg-GNN-NTD	4.1 s	61x	0.7	17.8 s	120x	2.5	19.1 s	147x	2.1
goc-2000	Org-NTD	>10.0 hr	-	-	>10.0 hr	-	-	>10.0 hr	-	-
	5-hop-NTD	3.4 hr	>3x	-9.2	>10.0 hr	>1x	-3.0	>10.0 hr	>1x	-6.7
	No-SW	5.5 s	>6510x	0.0	6.3 s	>5735x	0.0	14.0 s	>2573x	0.0
	Clf-GNN-NTD	20.0 s	>1799x	-4.3	38.3 s	>940x	-0.5	54.0 s	>666x	-2.8
	Reg-GNN-NTD	11.8 s	>3049x	-6.3	2.5 min	>244x	-1.1	1.3 min	>462x	-4.4

Conclusion and future work

- Takeaway message:
 - ✓ Candidate substation selection drastically reduces the search space
 - ✓ Topology-aware formulation and proximity filtering improve generalization
- Open challenges:
 - Zero-shot generalization to new systems
 - Combination of actions
 - Labeled training data

Ali Rajaei, Peter Palensky, Jochen L. Cremer,
“*Transferable Graph Learning for Transmission Congestion
Management via Busbar Splitting*,”
24th Power Systems Computation Conference (PSCC),
2026. ([Link](#))

Codes available on GitHub. ([Link](#))



Thank you for your attention!

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